Steam Leak Localization Algorithm

March 5, 2026

The goal is to determine the position of a steam leak with respect to the robot dog given the following measurements at the robot dog's current and the dog's previous locations:

Current $\boldsymbol{x}^D_{D/I_i} = \left[x^D_{D/I_i}, y^D_{D/I_i}\right]^T$: Translational position of the Dog Frame, D, with respect to the Inertial Frame (stationary observer), I

θ_{I/D_i}: Angular position of the Inertial Frame with respect to the Dog Frame

θ_{L/D_i}: Three speaker measurement of the leak direction with respect to the Dog Frame

Previous $\boldsymbol{x}^D_{D/I_{i-1}} = \left[x^D_{D/I_{i-1}}, y^D_{D/I_{i-1}}\right]^T$: Previous translational position of the Dog Frame with respect to the Inertial Frame

$\theta_{I/D_{i-1}}$: Previous angular position of the Inertial Frame with respect to the Dog Frame

$\theta_{L/D_{i-1}}$: Previous three speaker measurement of the leak direction with respect to the Dog Frame

In the above measurements and throughout this document, the subscripts i and $i-1$ denote the current and previous measurements taken by the dog, respectively. Figure 1 presents each of the four frames used throughout this derivation along with the repective position measurements.

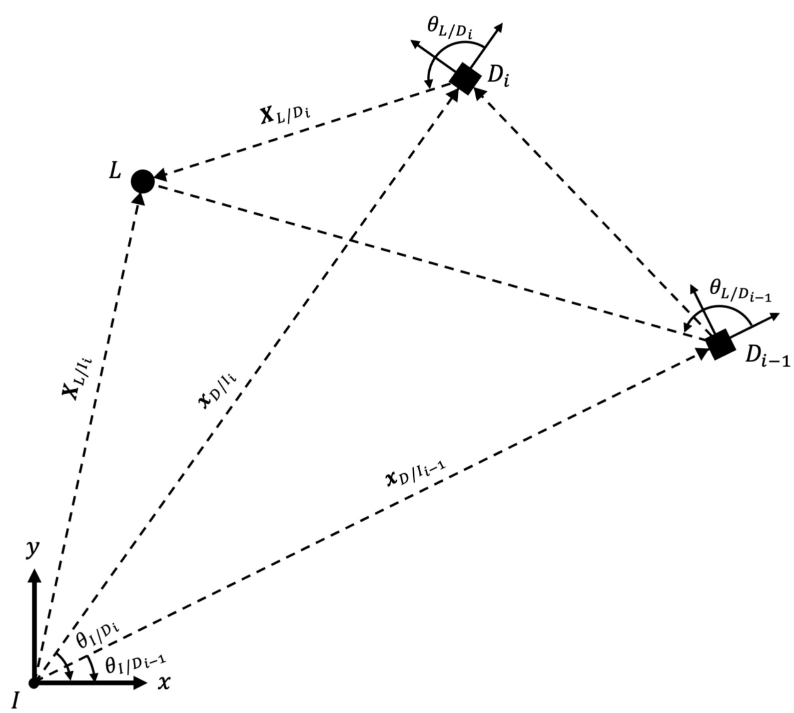

Figure 1: Reference Frames (I: Inertial Frame, L: Steam Leak, D_i: Dog Frame at the i^{th} (current) location, D_{i-1}: Dog Frame at the $i^{th}-1$ (previous) location)

Subtracting the angular position of the Inertial Frame with respect to the Dog Frame from the three speaker measurement of the leak direction with respect to the Dog Frame, provides the angular position of the leak with respect to the Inertial Frame,

$$\theta_{L/I_i} = \theta_{L/D_i} - \theta_{I/D_i} \tag{1}$$

To find the translational position of the dog with respect to the Inertial Frame resolved in the Inertial Frame, use the transformation:

$$\boldsymbol{x}^I_{D/I_i} = T_{I/D_i}\boldsymbol{x}^D_{D/I_i} \tag{2}$$

where $T_{I/D}$ is the rotation matrix,

$$T_{I/D_i} = \begin{bmatrix} \cos\left(\theta_{I/D_i}\right) & -\sin\left(\theta_{I/D_i}\right) \\ \sin\left(\theta_{I/D_i}\right) & \cos\left(\theta_{I/D_i}\right) \end{bmatrix} \tag{3}$$

The following equations will estimate the position of the steam leak with respect to a stationary observer, $\boldsymbol{X}^I_{L/I} = \left[X^I_{L/I}, Y^I_{L/I}\right]^T$:

$$X^I_{L/I} = \frac{y^I_{D/I_i} - y^I_{D/I_{i-1}} + \left|x^I_{D/I_i}\right| \tan\left(\theta_{L/I_i}\right) - \left|x^I_{D/I_{i-1}}\right| \tan\left(\theta_{L/I_{i-1}}\right)}{\tan\left(\theta_{L/I_{i-1}}\right) - \tan\left(\theta_{L/I_i}\right)} \tag{4}$$

$$Y^I_{L/I} = \tan\left(\theta_{L/I_{i-1}}\right) X^I_{L/I} + y^I_{D/I_{i-1}} + \left|x^I_{D/I_{i-1}}\right| \tan\left(\theta_{L/I_{i-1}}\right) \tag{5}$$

Finding the position of the leak with respect to the robot dog,

$$\boldsymbol{X}^I_{L/D_i} = \boldsymbol{X}^I_{L/I} - \boldsymbol{x}^I_{D/I_i} \tag{6}$$

If you want to resolve this in the Dog's Frame, simply use that following transformation,

$$\boldsymbol{X}^D_{L/D_i} = T^T_{I/D_i}\boldsymbol{X}^I_{L/D_i} \tag{7}$$

where the superscript T denotes the transpose of the corresponding matrix. After the current measurements, i, are taken make the dog walk in the direction θ_{L/D_i} for a distance of:

$$\text{Walking Distance} = \begin{cases} L_i & \text{if} \quad L_i < 1 \\ 1 & \text{if} \quad L_i \geq 1 \end{cases} \qquad \text{where,} \quad L_i = \frac{1}{10}\left\|\boldsymbol{X}^I_{L/D_i}\right\|_2 = \frac{1}{10}\left\|\boldsymbol{X}^D_{L/D_i}\right\|_2 \tag{8}$$

Here, $\|.\|_2$ denotes the Euclidean norm.

Geometry-Aware Selection of the Next Measurement Pose

In a room or corridor, the dog may not be able to move exactly 90° from the measured leak bearing. What improves the two-bearing geometry is the component of the motion from D_i to D_{i+1} that is perpendicular to the current leak-direction measurement θ_{L/D_i}.

Let θ_{D_{i+1}/D_i} denote a candidate travel heading from D_i to D_{i+1} in the current dog frame. Let L_i denote the traveled distance, and let $L_{\max,i}\left(\theta_{D_{i+1}/D_i}\right)$ denote the maximum collision-free distance available in that heading.

$$B_{\perp,i} = L_i \left| \sin\left(\theta_{D_{i+1}/D_i} - \theta_{L/D_i}\right) \right| \tag{9}$$

$$B_{\parallel,i} = L_i \cos\left(\theta_{D_{i+1}/D_i} - \theta_{L/D_i}\right) \tag{10}$$

Equation (9) is the effective perpendicular baseline. This component most strongly changes the second bearing; motion parallel to θ_{L/D_i} is much less useful.

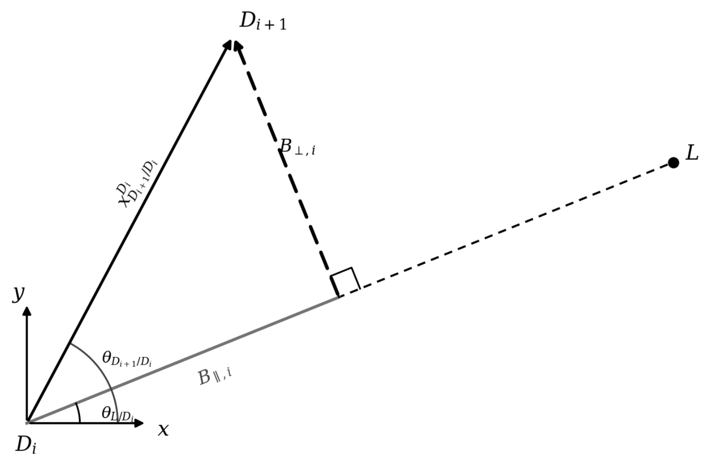

Figure 2: $B_{\perp,i}$ is the part of the motion from D_i to D_{i+1} that most improves the two-bearing geometry.

For a desired perpendicular baseline $B_{\perp,\mathrm{target}}$ (for example, 1 m), the required travel distance is

$$L_{i,\mathrm{req}} = \frac{B_{\perp,\mathrm{target}}}{\max\left(\left|\sin\left(\theta_{D_{i+1}/D_i} - \theta_{L/D_i}\right)\right|, \epsilon\right)} \tag{11}$$

where $\epsilon > 0$ is a small safeguard for nearly parallel motion.
Select the feasible heading that maximizes the usable perpendicular baseline:

$$\theta^{\star}_{D_{i+1}/D_i} = \mathrm{argmax}_{\theta_{D_{i+1}/D_i} \in \mathcal{F}_i} \left[L_{\max,i}\left(\theta_{D_{i+1}/D_i}\right) \left| \sin\left(\theta_{D_{i+1}/D_i} - \theta_{L/D_i}\right) \right| \right] \tag{12}$$

Here $\mathcal{F}_i$ is the set of collision-free headings returned by the local scan, occupancy grid, or local planner. After selecting $\theta^{\star}_{D_{i+1}/D_i}$, command the dog to move

$$L_{i,\mathrm{move}} = \min\left(L_{\max,i}\left(\theta^{\star}_{D_{i+1}/D_i}\right), \frac{B_{\perp,\mathrm{target}}}{\max\left(\left|\sin\left(\theta^{\star}_{D_{i+1}/D_i} - \theta_{L/D_i}\right)\right|, \epsilon\right)} \right) \tag{13}$$

After the motion, acquire the next leak-direction measurement at D_{i+1}. If the maximum achievable value in Eq. (12) is below a chosen minimum threshold, the geometry is weak; move farther if space allows, or take a third measurement before finalizing the leak estimate. This replaces the assumption in Eq. (8) that the dog can always walk in a single prescribed direction and instead balances localization geometry with local free-space constraints.